

Code-Layout, Readability and Reusability

Date: 17 july 2026
Team ID: XXXXXX
Project Name: Rising Waters: A Machine Learning Approach to Flood Prediction
Maximum Marks: 5 Marks

S.No	Code Quality Parameter	Description	Followed	Remarks
1	Consistent Indentation	Uniform spacing throughout code	Yes	PEP8 followed
2	Proper File Structure	Logical project folders	Yes	Flask MVC structure
3	Meaningful Variable Names	Clear descriptive names	Yes	Self-explanatory variables
4	Function/Method Names	Descriptive functions	Yes	Prediction & preprocessing modules
5	Code Comments	Inline/block comments	Yes	Major functions documented
6	Modular Design	Reusable modules	Yes	Separate preprocessing, training, prediction
7	No Redundant Code	Duplicate code removed	Yes	Reusable functions
8	Error Handling	Graceful exception handling	Partial	Input validation implemented

Reusable Components / Modules

Component	Technology	Where Reused	Level
Data Preprocessing	Python, Pandas	Training & Prediction	High
Feature Scaling	Scikit-learn	Training & Deployment	High
XGBoost Model	XGBoost	Prediction Engine	High
Flask Routes	Flask	Web Application	High
HTML Templates	HTML/CSS/Bootstrap	Frontend	Medium

Overall Code Quality Assessment

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5	Well-organized project structure
Readability	5	Clear naming conventions
Reusability	5	Modular reusable components
Documentation / Comments	4	Well documented
Overall Score	4.8/5	Production-ready academic project